

Practice Beginning Algebra Test III (Version 1)

Instructions

- As always, write a title, name, date, and course.
- Number each problem, circle the number, and show work when necessary.
- Draw a rectangle around your answers.

Solve.

$$\textcircled{1} \frac{\sqrt{x}}{3} + 2 = 7$$

$$\frac{\sqrt{x}}{3} = 5$$

$$\sqrt{x} = 15$$

$$(\sqrt{x})^2 = 15^2$$

$$\boxed{x = 225}$$

$$\textcircled{2} 10x^2 = 15x$$

$$10x^2 - 15x = 0$$

$$5x(2x - 3) = 0$$

$$5x = 0 \text{ or } 2x - 3 = 0$$

$$\boxed{x = 0} \text{ or } \boxed{x = \frac{3}{2}}$$

$$\textcircled{3} 32x = 7x^2 + 16$$

$$0 = 7x^2 - 32x + 16$$

$$0 = (7x - 4)(x - 4)$$

$$7x - 4 = 0 \text{ or } x - 4 = 0$$

$$\boxed{x = \frac{4}{7}} \text{ or } \boxed{x = 4}$$

$$\textcircled{4} x^3 - 4x - 4 = -x^2$$

$$x^3 + x^2 - 4x - 4 = 0$$

$$x^2(x + 1) - 4(x + 1) = 0$$

$$(x + 1)(x^2 - 4) = 0$$

$$(x + 1)(x - 2)(x + 2) = 0$$

$$x + 1 = 0 \text{ or } x - 2 = 0 \text{ or } x + 2 = 0$$

$$\boxed{x = -1} \text{ or } \boxed{x = 2} \text{ or } \boxed{x = -2}$$

$$\textcircled{5} \frac{\sqrt{y}}{-4} + 1 = 6$$

$$\frac{\sqrt{y}}{-4} = 5$$

$$\sqrt{y} = -20$$

No Solution

$$\textcircled{6} 2 = -5 + 4\sqrt{3 - \frac{x}{6}}$$

$$7 = 4\sqrt{3 - \frac{x}{6}}$$

$$\frac{7}{4} = \sqrt{3 - \frac{x}{6}}$$

$$\left(\frac{7}{4}\right)^2 = \left(\sqrt{3 - \frac{x}{6}}\right)^2$$

$$\frac{49}{16} = 3 - \frac{x}{6}$$

$$\frac{49}{16} - 3 = -\frac{x}{6}$$

$$\frac{1}{16} = -\frac{x}{6}$$

$$\boxed{-\frac{3}{8} = x}$$

$$\textcircled{7} \sqrt{x^2+7x} = \sqrt{18}$$

$$(\sqrt{x^2+7x})^2 = (\sqrt{18})^2$$

$$x^2+7x = 18$$

$$x^2+7x-18=0$$

$$(x+9)(x-2)=0$$

$$x+9=0 \text{ or } x-2=0$$

$$\boxed{x=-9} \text{ or } \boxed{x=2}$$

$$\textcircled{8} 15 = -3 + 7y^2$$

$$18 = 7y^2$$

$$\frac{18}{7} = y^2$$

$$\pm \sqrt{\frac{18}{7}} = \sqrt{y^2}$$

$$\pm \frac{\sqrt{18}}{\sqrt{7}} = y$$

$$\pm \frac{3\sqrt{2}}{\sqrt{7}} = y$$

$$\boxed{\pm \frac{3\sqrt{14}}{7} = y}$$

$$\textcircled{9} 0 = 12x^2 - 27x$$

$$0 = 3x(4x-9)$$

$$3x=0 \text{ or } 4x-9=0$$

$$\boxed{x=0} \text{ or } \boxed{x=\frac{9}{4}}$$

$$\textcircled{10} -6 = 2\left(\frac{x}{4}+5\right)^2 - 8$$

$$2 = 2\left(\frac{x}{4}+5\right)^2$$

$$1 = \left(\frac{x}{4}+5\right)^2$$

$$\pm \sqrt{1} = \sqrt{\left(\frac{x}{4}+5\right)^2}$$

$$\pm 1 = \frac{x}{4}+5$$

$$1 = \frac{x}{4}+5 \text{ or } -1 = \frac{x}{4}+5$$

$$\boxed{-16=x} \text{ or } \boxed{-24=x}$$

$$\textcircled{11} \quad 2x-1 = \frac{3x}{5} - \frac{3}{10}$$

$$\frac{2x}{1} - \frac{1}{1} = \frac{3x}{5} - \frac{3}{10}$$

$$\frac{20x}{10} - \frac{10}{10} = \frac{6x}{10} - \frac{3}{10}$$

$$\frac{20x-10}{10} = \frac{6x-3}{10}$$

$$20x-10 = 6x-3$$

$$14x = 7$$

$$\boxed{x = \frac{1}{2}}$$

$$\textcircled{12} \quad 10x^5 - 40x^3 = 210x$$

$$10x^5 - 40x^3 - 210x = 0$$

$$10x(x^4 - 4x^2 - 21) = 0$$

$$10x(x^2-7)(x^2+3) = 0$$

$$10x=0 \text{ or } x^2-7=0 \text{ or } x^2+3=0$$

$$\boxed{x=0} \quad x^2=7 \quad x^2 \neq -3$$

$$\sqrt{x^2} = \pm\sqrt{7}$$

$$\boxed{x = \pm\sqrt{7}}$$

$$\textcircled{13} \quad x+10 = -\frac{25}{x}$$

$$\frac{x}{1} + \frac{10}{1} = -\frac{25}{x}$$

$$\frac{x^2}{x} + \frac{10x}{x} = -\frac{25}{x}$$

$$\frac{x^2+10x}{x} = -\frac{25}{x}$$

$$x^2+10x = -25$$

$$x^2+10x+25 = 0$$

$$(x+5)(x+5) = 0$$

$$x+5 = 0$$

$$\boxed{x = -5}$$

$$\textcircled{14} \quad (7-2x)^2 + 2 = 7$$

$$(7-2x)^2 = 5$$

$$\sqrt{(7-2x)^2} = \pm\sqrt{5}$$

$$7-2x = \pm\sqrt{5}$$

$$-2x = \pm\sqrt{5}-7$$

$$x = \frac{\pm\sqrt{5}-7}{-2}$$

$$\boxed{x = \frac{\pm\sqrt{5}+7}{2}}$$

$$\textcircled{15} \frac{11}{3y^3} + \frac{8}{y^4} = -\frac{1}{3y^2}$$

$$\frac{11 \cdot y}{3y^3 \cdot y} + \frac{8 \cdot 3}{y^4 \cdot 3} = -\frac{1 \cdot y^2}{3y^2 \cdot y^2}$$

$$\frac{11y}{3y^4} + \frac{24}{3y^4} = -\frac{y^2}{3y^4}$$

$$\frac{11y+24}{3y^4} = -\frac{y^2}{3y^4}$$

$$11y+24 = -y^2$$

$$y^2+11y+24=0$$

$$(y+8)(y+3)=0$$

$$y+8=0 \text{ or } y+3=0$$

$$\boxed{y=-8} \text{ or } \boxed{y=-3}$$

$$\textcircled{17} -56x^3+192x^2=-4x^4$$

$$4x^4-56x^3+192x^2=0$$

$$4x^2(x^2-14x+48)=0$$

$$4x^2(x-6)(x-8)=0$$

$$4x^2=0 \text{ or } x-6=0 \text{ or } x-8=0$$

$$\boxed{x=0} \text{ or } \boxed{x=6} \text{ or } \boxed{x=8}$$

$$\textcircled{16} -\frac{1}{8} - \frac{5}{x} = 2 - \frac{3}{8x}$$

$$-\frac{1}{8} - \frac{5}{x} = \frac{2}{1} - \frac{3}{8x}$$

$$-\frac{x}{8x} - \frac{40}{8x} = \frac{16x}{8x} - \frac{3}{8x}$$

$$\frac{-x-40}{8x} = \frac{16x-3}{8x}$$

$$-x-40=16x-3$$

$$-37=17x$$

$$\boxed{\frac{-37}{17}=x}$$

$$\textcircled{18} x^3+4x^2+x+4=0$$

$$x^2(x+4)+1(x+4)=0$$

$$(x+4)(x^2+1)=0$$

$$x+4=0 \text{ or } x^2+1=0$$

$$\boxed{x=-4} \quad x^2 \neq -1$$

$$\textcircled{19} 4b^3-36b^2=11b-99$$

$$4b^3-36b^2-11b+99=0$$

$$4b^2(b-9)-11(b-9)=0$$

$$(b-9)(4b^2-11)=0$$

$$b-9=0 \text{ or } 4b^2-11=0$$

$$\boxed{b=9} \text{ or } 4b^2=11$$

$$b^2=\frac{11}{4}$$

$$\sqrt{b^2}=\pm\sqrt{\frac{11}{4}}$$

$$\boxed{b=\pm\frac{\sqrt{11}}{2}}$$

$$\textcircled{20} \frac{7(9x-2)^2}{5} + 3 = 6$$

$$\frac{7(9x-2)^2}{5} = 3$$

$$7(9x-2)^2 = 15$$

$$(9x-2)^2 = \frac{15}{7}$$

$$\sqrt{(9x-2)^2} = \pm\sqrt{\frac{15}{7}}$$

$$9x-2 = \pm\sqrt{\frac{15}{7}}$$

$$9x-2 = \pm\frac{\sqrt{105}}{7}$$

$$9x = 2 \pm \frac{\sqrt{105}}{7}$$

$$\boxed{x = \frac{2 \pm \frac{\sqrt{105}}{7}}{9}}$$

$$x = \frac{14 \pm \sqrt{105}}{63}$$

$$(21) \frac{3}{x-6} + \frac{1}{x+1} = \frac{2x+9}{x^2-5x-6}$$

$$\frac{3}{x-6} + \frac{1}{x+1} = \frac{2x+9}{(x+1)(x-6)}$$

$$\frac{3(x+1)}{(x-6)(x+1)} + \frac{1(x-6)}{(x+1)(x-6)} = \frac{2x+9}{(x+1)(x-6)}$$

$$\frac{3x+3}{(x-6)(x+1)} + \frac{x-6}{(x+1)(x-6)} = \frac{2x+9}{(x+1)(x-6)}$$

$$\frac{4x-3}{(x-6)(x+1)} = \frac{2x+9}{(x+1)(x-6)}$$

$$4x-3 = 2x+9$$

$$2x = 12$$

$$x \neq 6$$

No Solution

$$(22) \frac{-1}{x+2} - \frac{3}{x^2+2x} = \frac{1}{x}$$

$$\frac{-1}{x+2} - \frac{3}{x(x+2)} = \frac{1}{x}$$

$$\frac{-1 \cdot x}{(x+2)x} - \frac{3}{x(x+2)} = \frac{1(x+2)}{x(x+2)}$$

$$\frac{-x}{x(x+2)} - \frac{3}{x(x+2)} = \frac{x+2}{x(x+2)}$$

$$\frac{-x-3}{x(x+2)} = \frac{x+2}{x(x+2)}$$

$$-x-3 = x+2$$

$$-5 = 2x$$

$$\boxed{-\frac{5}{2} = x}$$

$$(23) \frac{3}{x^2+7x+12} = \frac{1}{x+4} - \frac{x+1}{x+3}$$

$$\frac{3}{(x+4)(x+3)} = \frac{1}{x+4} - \frac{x+1}{x+3}$$

$$\frac{3}{(x+4)(x+3)} = \frac{1(x+3)}{(x+4)(x+3)} - \frac{(x+1)(x+4)}{(x+3)(x+4)}$$

$$\frac{3}{(x+4)(x+3)} = \frac{x+3}{(x+4)(x+3)} - \frac{x^2+5x+4}{(x+3)(x+4)}$$

$$\frac{3}{(x+4)(x+3)} = \frac{x+3-x^2-5x-4}{(x+4)(x+3)}$$

$$\frac{3}{(x+4)(x+3)} = \frac{-x^2-4x-1}{(x+4)(x+3)}$$

$$3 = -x^2-4x-1$$

$$x^2+4x+4=0$$

$$(x+2)(x+2)=0$$

$$x+2=0 \text{ or } x+2=0$$

$$\boxed{x=-2}$$

$$(24) x-1 = \sqrt{-4x+9}$$

$$(x-1)^2 = (\sqrt{-4x+9})^2$$

$$(x-1)(x-1) = -4x+9$$

$$x^2 - x - x + 1 = -4x + 9$$

$$x^2 - 2x + 1 = -4x + 9$$

$$x^2 + 2x - 8 = 0$$

$$(x+4)(x-2) = 0$$

$$x+4=0 \text{ or } x-2=0$$

$$x \neq -4 \text{ or } \boxed{x=2}$$

$$(25) \sqrt{x}-2 = \sqrt{x-8}$$

$$(\sqrt{x}-2)^2 = (\sqrt{x-8})^2$$

$$(\sqrt{x}-2)(\sqrt{x}-2) = x-8$$

$$\sqrt{x} \cdot \sqrt{x} - 2\sqrt{x} - 2\sqrt{x} + 4 = x-8$$

$$\sqrt{x^2} - 4\sqrt{x} + 4 = x-8$$

$$x - 4\sqrt{x} + 4 = x-8$$

$$-4\sqrt{x} + 4 = -8$$

$$-4\sqrt{x} = -12$$

$$\sqrt{x} = 3$$

$$(\sqrt{x})^2 = 3^2$$

$$\boxed{x=9}$$